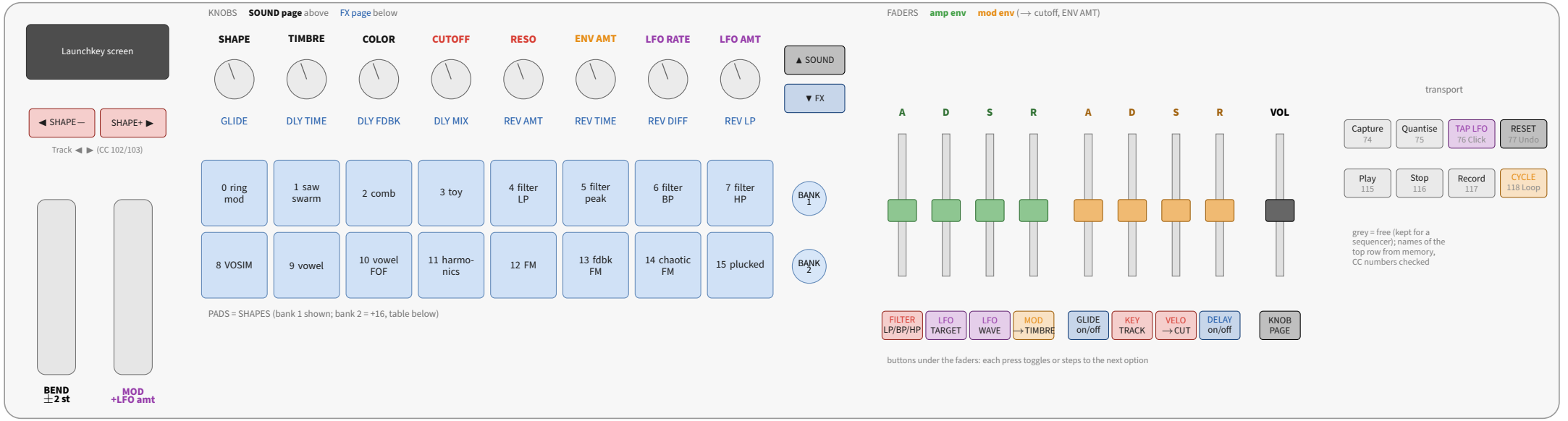
freak

Axoloti Core + Launchkey 49 MK3 (standalone mode) · 3-voice Braids synth · patch `patches/freak/freak_v1.axp`



Knobs: two pages, scaled pickup

SHAPE	34 Braids shapes in turn (3–4 steps each)
TIMBRE / COLOR	the two Braids parameters (meaning depends on the shape)
CUTOFF	20 Hz $\cdot 2^{10k}$: 20 Hz – 20 kHz
RESO	Q = 0.5 – 20 (exponential)
ENV AMT	mod env → cutoff, ±5 octaves, 0 at centre
LFO RATE	0.05 – 20 Hz (or tap tempo)
LFO AMT	0 – 1, plus the mod wheel
GLIDE	0 – 1 s (quadratic), with GLIDE on
DLY TIME / FDBK / MIX	10 ms – 1.5 s; 0 – 0.95; 0 – 1
REV AMT/TIME/DIFF/LP	Elements reverb, 0 – 1 each

Page: ▲/▼ by the knobs, or the 9th fader button (the page is not shown!). After a page change or RESET, a knob that is not where its value is still works: the value moves *proportionally* the way you turn and meets the knob at the latest at an end of travel. No jumps, no dead knobs. On the FX page, knobs 1–4 are silent until GLIDE or DELAY is switched on.

Envelopes and filter

Faders 1–4 **amp**, 5–8 **mod**: A, D, R = 1 ms $\cdot 10^{4k}$ (1 ms – 10 s, a decade per quarter of the travel); S = level 0 – 1. Exponential segments; attack aims at 1.2: a concave rise, fast at first (analog style).

Mod env → cutoff (ENV AMT), and → timbre (±0.5) with MOD→TIMBRE. **CYCLE** (Loop): it loops attack/decay while a key is held.

Filter: 12 dB/oct state-variable (LP/BP/HP, BP normalised). KEY TRACK (on at power-on): cutoff follows the keys, 1 oct/oct around note 60 (middle C). VELO→CUT: up to +2 octaves for a hard hit.

LFO (one, shared by the voices)

TARGET: **timbre** → color → cutoff (±3 oct) → pitch (±2 st). WAVE: **sine** → triangle → saw → square → S&H. TAP LFO: two taps 0.1–4 s apart set the period; LFO RATE takes over again when turned.

Pads: shapes, two banks

Bank 1 (0–15)
0 ring mod 1 saw swarm
2 comb 3 toy
4–7 filter LP/peak/BP/HP
8 VOSIM 9 vowel
10 vowel FOF 11 harmonics
12 FM 13 feedback FM
14 chaotic FM 15 plucked

Bank 2 (16–31)
16 bowed 17 blown
18 fluted 19 bell
20 drum 21 kick
22 cymbal 23 snare
24 wavetables 25 wave map
26 wave line 27 paraphonic
28 filtered noise 29 twin peaks
30 clocked noise 31 granular

Top row = first 8, bottom row = next 8. 32 particle noise and 33 digital modulation: ▶ from 31, or the SHAPE knob. Strike-type shapes (plucked, bell, drums) are struck on each key press.

Notes

3 voices. Power-on: saw swarm, LP, key tracking on, delay off. RESET (Undo) restores that sound and keeps the page. The buttons' states are **not shown** (no LED feedback yet). Every press moves to the next option, so count, or listen.

The Launchkey's own features

They make notes, so they play freak as well (set them on the Launchkey, see its user guide):

Arp: up, down, up/down, as played, random, chord, mutate; 1/4 – 1/32T; 1–4 oct; latch, gate, swing, rhythm.
Scale: minor, major, **Dorian**, **Mixolydian**, Phrygian, harmonic minor, minor/major pentatonic.
Fixed chord, octave ±, transpose.
Not yet tested with freak.

Signal path and load

3 × (Braids → SVF → amp env × velocity) → delay → Elements reverb → VOL. Velocity: soft notes keep $\frac{1}{4}$ of the level. DSP 55 %, patch RAM 41 of 44 KB (Axoloti Core, Ksoloti 1.1.0).